Arista Access Node KR-ACC-24 Optical Transport Failure SOP (ALARM_1685)

Node: KR-ACC-24 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1685

Description

This Standard Operating Procedure (SOP) addresses the Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Arista Access Node KR-ACC-24. The failure domain is primarily caused by the degradation of the SFP/XFP transceivers, which are used for optical transmission between the node and the optical transport network. This degradation can occur due to various reasons such as aging, environmental factors, or physical damage. In some cases, fiber cuts or Rx/Tx power loss can also cause the alarm to be raised. On Arista Access Nodes, the optical transport failure is detected by the hardware and reported as an alarm (ALARM_1685) with a severity of Warning. This SOP outlines the diagnostic and resolution steps to address the issue, and ensures that the node is restored to a stable state.

Pre-requisite Checks

- Check: Verify the redundant link status on KR-ACC-24 to ensure that the failure does not impact the network redundancy., Description: The redundant link status should be checked to ensure that the failure does not impact the network redundancy.

- Check: Confirm that the remote hands have been dispatched to KR Puram Access to assist with the troubleshooting and repair process., Description: The remote hands should be dispatched to KR Puram Access to assist with the troubleshooting and repair process.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware transceiver all, Expected Output: Transceiver information for all SFP/XFP slots, Description: This command displays the transceiver information for all SFP/XFP slots on the node, including the transceiver type, serial number, and status.

Command: show hardware optics all, Expected Output: Optics information for all interfaces, Description: This command displays the optics information for all interfaces on the node, including the interface type, speed, and status.
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Command: show optics stats, Expected Output: Optics statistics for all interfaces, Description: This command displays the optics statistics for all interfaces on the node, including the received and transmitted bytes, packets, and errors.
```

Resolution Steps

Step 1: Step: Replace the failed SFP/XFP transceiver, Command: hw-module transceiver replace slot , Description: This command replaces the failed SFP/XFP transceiver in the specified slot with a new one.

Step 2: Step: Verify the optics status, Command: show hardware optics all, Description: This command verifies the optics status for all interfaces on the node to ensure that the issue has been resolved.

Step 3: Step: Clear the alarm, Command: clear alarm ALARM_1685, Description: This command clears the alarm (ALARM_1685) to indicate that the issue has been resolved.

Post-Check Verification

```
Command: show hardware transceiver all, Expected Output: Transceiver information for all SFP/XFP slots, Description: This command verifies that the transceiver information for all SFP/XFP slots on the node is accurate and up-to-date.

Command: show hardware optics all, Expected Output: Optics information for all interfaces, Description: This command verifies that the optics information for all interfaces on the node is accurate and up-to-date.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: hw-module transceiver restore slot , Description: This command restores the original SFP/XFP transceiver in the specified slot.

- Step: Clear the alarm (if necessary), Command: clear alarm ALARM_1685, Description: This command clears the alarm (ALARM_1685) if necessary to indicate that the original state has been restored.